Chapter 1

Linear classification: A Summary

Introduction to Classification

- Classification predicts discrete outcomes (class labels) rather than continuous values.
- Input space partitioned into **decision regions**, each corresponding to a class.
- **Linear classification models**: decision boundaries are hyperplanes in feature space.
- Datasets separable by such boundaries are **linearly separable**.
- Target representation:
  - Binary: $t \in \{0, 1\}$.
  - Multiclass ($K > 2$): **1-of-K coding** (vector of K bits, exactly one 1).

Three Approaches to Classification

1. **Discriminant function**: direct mapping $f : \mathbf{X} \mapsto \{1, \dots, K\}$.
2. **Discriminative approach**: infer $p(C_j|\mathbf{x})$ (inference), then assign class (decision).
3. **Generative approach**: model $p(\mathbf{x}|C_j)$ and $p(C_j)$, derive posteriors via Bayes.

- Approaches 1 and 2 are **discriminative** (directly tackle classification).
- For classification, adapt linear regression via **generalized linear model**:

$$h(\mathbf{x}; \mathbf{w}, b) = f(\mathbf{w}^T\mathbf{x} + b) \tag{1.1}$$

  with f an **activation function** mapping to $[0, 1]$.
- Decision boundaries: $\mathbf{w}^T\mathbf{x} + b = f^{-1}(c)$ (linear in feature space). f^{-1} is **link function**.
- Generative approach: define probabilistic models for items within each class based on training data.

Binary Classification with Linear Discriminants

- Equation $\mathbf{w}^T\mathbf{x} + b = 0$ defines a hyperplane in $\mathbb{R}^d$.
- For $\mathbf{x}_0$ on hyperplane: $\mathbf{w}^T\mathbf{x}_0 + b = 0 \implies b = -\mathbf{w}^T\mathbf{x}_0$, giving $\mathbf{w}^T(\mathbf{x} - \mathbf{x}_0) = 0$.
- $\mathbf{w}$ is a normal vector (perpendicular to every vector lying in the hyperplane).
- $\mathbf{w}^T\mathbf{x}$ measures alignment with $\mathbf{w}$; hyperplane is level set of this linear functional.
- Normalized form: $\frac{\mathbf{w}^T}{\|\mathbf{w}\|}\mathbf{x} + \frac{b}{\|\mathbf{w}\|} = 0$, where $\frac{\mathbf{w}}{\|\mathbf{w}\|}$ is unit normal.
- $\frac{|b|}{\|\mathbf{w}\|}$ is perpendicular distance from origin to hyperplane.

- Sign of $\mathbf{w}^T\mathbf{x} + b$ indicates which side: positive $\Rightarrow$ side $\mathbf{w}$ points to, negative $\Rightarrow$ opposite.
- Magnitude $|\mathbf{w}^T\mathbf{x} + b|$ measures distance from decision boundary.
- Bias b shifts hyperplane along direction of $\mathbf{w}$.
- For classification with $t \in \{-1, +1\}$: correct classification when $t(\mathbf{w}^T\mathbf{x} + b) > 0$ (point on correct side with margin).
- Magnitude $|t(\mathbf{w}^T\mathbf{x} + b)|$ measures classification confidence.

figures/hyperplane.png

Figure 1.1. Vector view of hyperplane equation.

Multiclass Extension

One-Versus-Rest

- Train $K - 1$ binary classifiers f_i, each separating C_i from all other classes.
- Decision: if $f_i(\mathbf{x}) > 0$, assign to C_i; else continue.
- **Limitation**: Regions may overlap (multiple classifiers positive) or leave gaps (all classifiers negative).

One-Versus-One

- Train $K(K-1)/2$ classifiers f_{ij} for each pair (C_i, C_j).
- Decision by majority voting: each classifier votes for one class.
- **Limitations**: Regions may remain unassigned (ties), $O(K^2)$ complexity, possibility of voting cycles.

Multiclass Discriminant (Argmax)

- Define K linear functions $y_i(\mathbf{x}) = \mathbf{w}_i^T\mathbf{x} + b_i$.

figures/hyperplane1.png

Figure 1.2. Positive and negative hyperspaces.

- Classification rule: $k = \arg\max j y_j(\mathbf{x})$ (winner-take-all).
- Decision boundary between C_i and C_j: $y_i(\mathbf{x}) = y_j(\mathbf{x}) \implies (\mathbf{w}_i - \mathbf{w}_j)^T\mathbf{x} + (b_i - b_j) = 0$ (hyperplane).
- This partition is complete and unambiguous.
- Decision regions are **convex and connected**. Proof: if $\mathbf{x}_A, \mathbf{x}_B \in \mathcal{R}_k$, then for any convex combination $\hat{\mathbf{x}} = \lambda\mathbf{x}_A + (1-\lambda)\mathbf{x}_B$, $y_k(\hat{\mathbf{x}}) = \lambda y_k(\mathbf{x}_A) + (1-\lambda)y_k(\mathbf{x}_B) > \lambda y_j(\mathbf{x}_A) + (1-\lambda)y_j(\mathbf{x}_B) = y_j(\hat{\mathbf{x}})$ for all $j \neq k$.
- Convexity ensures stability under small perturbations.
- Trade-off: unified framework requires learning all K weight vectors jointly.

Beyond Linearity: Quadratic Discriminants

- Incorporate pairwise feature interactions:

$$h(\mathbf{x}; \mathbf{w}, b) = b + \sum_{i=1}^{d} w_i x_i + \sum_{i=1}^{d} \sum_{j=1}^{i} w_{ij} x_i x_j \tag{1.2}$$

- Adds $\frac{d(d+1)}{2}$ new parameters, enabling curved decision boundaries.
- Can be seen as linear classifier in expanded feature space (original features + pairwise products).

Least Squares Applied to Classification

- Use 1-of-K coding: $z_1, \dots, z_K$ with $z_i = 1$ for class C_i.
- Derive K linear regression functions $h_i(\mathbf{x}) = \mathbf{w}_i^T\mathbf{x} + b_i$ with z_i as targets.

figures/hyperplane2.png

Figure 1.3. Normalized distances.

- Classification: assign $\mathbf{x}$ to class C_k where $k = \underset{i}{\operatorname{argmax}}\, h_i(\mathbf{x})$.
- $h_i(\mathbf{x})$ estimates $\mathbb{E}[z_i|\mathbf{x}]$, which under Bernoulli assumption equals $P(C_i|\mathbf{x})$ (but not constrained to $[0, 1]$).
- Training set $(\mathbf{X}, \mathbf{t})$ with $\mathbf{x}_i \in \mathbb{R}^d$, $\mathbf{t}_i \in \{0, 1\}^K$.
- Predictions: $\mathbf{y}_i = \mathbf{h}(\mathbf{x}_i) = \mathbf{W}\mathbf{x}_i + \mathbf{b}$.
- Augmented parameter matrix $\overline{\mathbf{W}}$ (including biases) and design matrix $\overline{\mathbf{X}}$:

$$\overline{\mathbf{W}} = \begin{pmatrix} w_{10} & w_{11} & \cdots & w_{1d} \\ w_{20} & w_{21} & \cdots & w_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ w_{K0} & w_{K1} & \cdots & w_{Kd} \end{pmatrix}, \quad \overline{\mathbf{X}} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1d} \\ 1 & x_{21} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nd} \end{pmatrix} \tag{1.3}$$

- Predictions matrix: $\overline{\mathbf{Y}} = \overline{\mathbf{W}}\overline{\mathbf{X}}^T$.
- Target matrix $\mathbf{T}$ $(K \times n)$ with $\mathbf{T}_{ij} = t_{ji}$.
- Residual matrix $\mathbf{R} = \overline{\mathbf{W}}\overline{\mathbf{X}}^T - \mathbf{T}$.
- Sum of squared residuals = $\operatorname{tr}(\mathbf{R}^T\mathbf{R})$.
- Objective: $E(\overline{\mathbf{W}}) = \frac{1}{2}\operatorname{tr}(\mathbf{R}^T\mathbf{R})$.
- Gradient: $\nabla_{\mathbf{W}} E(\overline{\mathbf{W}}) = \overline{\mathbf{X}}^T\overline{\mathbf{X}}\overline{\mathbf{W}} - \overline{\mathbf{X}}^T\mathbf{T}$.
- Closed-form solution:

$$\overline{\mathbf{W}} = (\overline{\mathbf{X}}^T\overline{\mathbf{X}})^{-1}\overline{\mathbf{X}}^T\mathbf{T} \tag{1.4}$$

- Discriminant functions:

$$\mathbf{h}(\mathbf{x}) = \overline{\mathbf{W}}^T \overline{\mathbf{x}} = \mathbf{T}^T \overline{\mathbf{X}} (\overline{\mathbf{X}}^T \overline{\mathbf{X}})^{-1} \overline{\mathbf{x}} \tag{1.5}$$

- **Limitations**: Predictions not constrained to $[0, 1]$, sensitive to outliers and class imbalance.

Fisher Linear Discriminant: Maximizing Class Separability

- Projects data onto lower-dimensional subspace to maximize class separation.
- Binary case: project onto direction $\mathbf{w}$: $y = \mathbf{w}^T \mathbf{x}$ (assume $\|\mathbf{w}\| = 1$).
- Classification: choose threshold $\tilde{y}$, assign $\mathbf{x}$ to C_1 if $y > \tilde{y}$.

Deriving Optimal Projection for Binary Classification

- Class means: $\mathbf{m}_1 = \frac{1}{n_1} \sum_{\mathbf{x} \in C_1} \mathbf{x}$, $\mathbf{m}_2 = \frac{1}{n_2} \sum_{\mathbf{x} \in C_2} \mathbf{x}$.
- Projected means: $m_i = \mathbf{w}^T \mathbf{m}_i$.
- Maximizing $m_2 - m_1$ alone gives $\mathbf{w} = \frac{\mathbf{m}_2 - \mathbf{m}_1}{\|\mathbf{m}_2 - \mathbf{m}_1\|}$.
- **Limitation**: ignores within-class spread; classes may overlap in projection.
- Within-class variance for class i: $s_i^2 = \sum_{\mathbf{x} \in C_i} (\mathbf{w}^T \mathbf{x} - m_i)^2$.
- Fisher criterion:

$$J(\mathbf{w}) = \frac{(m_2 - m_1)^2}{s_1^2 + s_2^2} \tag{1.6}$$

  Maximizes separation while minimizing dispersion.
- Within-class covariance matrix: $\mathbf{S}_i = \sum_{\mathbf{x} \in C_i} (\mathbf{x} - \mathbf{m}_i)(\mathbf{x} - \mathbf{m}_i)^T$.
- Projected variance: $s_i^2 = \mathbf{w}^T \mathbf{S}_i \mathbf{w}$.
- Total within-class covariance: $\mathbf{S}_W = \mathbf{S}_1 + \mathbf{S}_2$.
- Between-class covariance: $\mathbf{S}_B = (\mathbf{m}_2 - \mathbf{m}_1)(\mathbf{m}_2 - \mathbf{m}_1)^T$.
- Fisher criterion in matrix form:

$$J(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_W \mathbf{w}} \tag{1.7}$$

- Maximization leads to generalized eigenvalue problem:

$$\mathbf{S}_B \mathbf{w} = \lambda \mathbf{S}_W \mathbf{w} \tag{1.8}$$

- Solution:

$$\hat{\mathbf{w}} = \mathbf{S}_W^{-1} (\mathbf{m}_2 - \mathbf{m}_1) \tag{1.9}$$

- $\mathbf{S}_W^{-1}$ acts as whitening transform, down-weighting directions of high variance.

Determining the Optimal Threshold

- Assume projected values for each class are Gaussian.
- Estimate m_i, σ_i^2 via MLE.
- Bayes posterior: $p(C_i|y) \propto p(y|C_i)p(C_i) \propto n_i e^{-\frac{(y-m_i)^2}{2\sigma_i^2}}$.
- Optimal threshold where $p(C_1|y) = p(C_2|y)$.

Extending LDA to Multiclass Classification

- For $K > 2$, find D' projection directions ($1 < D' < D$): $y_k = \mathbf{w}_k^T \mathbf{x}$.
- Projection matrix $\mathbf{W}$ ($D \times D'$) gives $\mathbf{y} = \mathbf{W}^T \mathbf{x}$.
- Within-class scatter:

$$\mathbf{S}_W = \sum_{i=1}^{K} \sum_{\mathbf{x} \in C_i} (\mathbf{x} - \mathbf{m}_i)(\mathbf{x} - \mathbf{m}_i)^T \tag{1.10}$$

- Total covariance:

$$\mathbf{S}_T = \sum_{\mathbf{x}} (\mathbf{x} - \mathbf{m})(\mathbf{x} - \mathbf{m})^T = \mathbf{S}_W + \sum_{i=1}^{K} n_i (\mathbf{m}_i - \mathbf{m})(\mathbf{m}_i - \mathbf{m})^T \tag{1.11}$$

- Between-class scatter:

$$\mathbf{S}_B = \sum_{i=1}^{K} n_i (\mathbf{m}_i - \mathbf{m})(\mathbf{m}_i - \mathbf{m})^T \tag{1.12}$$

- Projected scatters: $\mathbf{s}_W = \mathbf{W}^T \mathbf{S}_W \mathbf{W}$, $\mathbf{s}_B = \mathbf{W}^T \mathbf{S}_B \mathbf{W}$.
- Objective functions:
  - Determinant ratio: $J_1(\mathbf{W}) = \frac{|\mathbf{s}_B|}{|\mathbf{s}_W|} = |(\mathbf{W}^T \mathbf{S}_W \mathbf{W})^{-1} \mathbf{W}^T \mathbf{S}_B \mathbf{W}|$.
  - Trace ratio: $J_2(\mathbf{W}) = \mathrm{tr}(\mathbf{s}_W^{-1} \mathbf{s}_B) = \mathrm{tr}(\mathbf{W}^T \mathbf{S}_W \mathbf{W})^{-1} \mathbf{W}^T \mathbf{S}_B \mathbf{W}$.
- Solutions via generalized eigenvalue problem: $\mathbf{S}_B \mathbf{w}_i = \lambda_i \mathbf{S}_W \mathbf{w}_i$.
- $\mathbf{W}^*$ consists of eigenvectors corresponding to D' largest eigenvalues.
- Rank of $\mathbf{S}_B$ at most $\min(K-1, D)$, so maximum meaningful $D' = K - 1$.
- Steps:
  1. Compute $\mathbf{S}_W$, $\mathbf{S}_B$.
  2. Solve $\mathbf{S}_B \mathbf{w} = \lambda \mathbf{S}_W \mathbf{w}$.
  3. Select top $K - 1$ eigenvectors.
  4. Form $\mathbf{W}$ and project data.
- **Limitations**:
  - $\mathbf{S}_W$ singular if $n < D$ (need regularization or PCA preprocessing).
  - Assumes Gaussian classes with equal covariance.

- Linear boundaries only.
- Sensitive to class imbalance (weighting by n_i).

- Useful for visualization: project onto first 2 or 3 discriminant directions.

Perceptron

- Foundational neural network model (1960s).
- Binary classifier with linear decision boundary:

$$h(\mathbf{x}) = f(\mathbf{w}^T\mathbf{x} + b) = f(\overline{\mathbf{w}}^T\overline{\mathbf{x}}), \quad f(i) = \begin{cases} -1 & i < 0 \\ 1 & i \geq 0 \end{cases} \tag{1.13}$$

- Targets $t_i \in \{-1, 1\}$.
- Can only classify linearly separable data.
- Cost function based on misclassified points:

$$E_p(\overline{\mathbf{w}}) = -\sum_{\mathbf{x}_i \in \mathcal{M}} \overline{\mathbf{w}}^T\overline{\mathbf{x}}_i t_i \tag{1.14}$$

  where $\mathcal{M}$ is set of misclassified points.
- Gradient: $\nabla_{\overline{\mathbf{w}}} E_p(\overline{\mathbf{w}}) = -\sum_{\mathbf{x}_i \in \mathcal{M}} \overline{\mathbf{x}}_i t_i$.
- Batch gradient descent:

$$\overline{\mathbf{w}}^{(k+1)} = \overline{\mathbf{w}}^{(k)} + \eta \sum_{\mathbf{x}_i \in \mathcal{M}_k} \overline{\mathbf{x}}_i t_i \tag{1.15}$$

- Online (stochastic) version:

$$\overline{\mathbf{w}}^{(k+1)} = \overline{\mathbf{w}}^{(k)} + \eta \overline{\mathbf{x}}_i t_i \tag{1.16}$$

  for a single misclassified $\mathbf{x}_i$.
- For a misclassified point, its cost decreases after update:

$$\begin{aligned} -\mathbf{x}_i^T\mathbf{w}^{(k+1)}t_i &= -\mathbf{x}_i^T\mathbf{w}^{(k)}t_i - \eta(\mathbf{x}_i t_i)^T\mathbf{x}_i t_i \\ &= -\mathbf{x}_i^T\mathbf{w}^{(k)}t_i - \eta\|\mathbf{x}_i\|^2 < -\mathbf{x}_i^T\mathbf{w}^{(k)}t_i \end{aligned}$$

- Correcting one point may cause another to become misclassified, but algorithm guaranteed to converge if data is linearly separable.

Convergence of the Perceptron Algorithm

- **Assumption**: Data linearly separable: $\exists \overline{\mathbf{w}}^*$ with $\|\overline{\mathbf{w}}^*\| = 1$ and $\overline{\mathbf{w}}^{*T}\overline{\mathbf{x}}_i t_i > 0$ for all i.
- **Margin**: $\gamma = \min_i \overline{\mathbf{w}}^{*T}\overline{\mathbf{x}}_i t_i > 0$.
- **Radius**: $R = \max_i \|\overline{\mathbf{x}}_i\|$.
- Consider $\overline{\mathbf{w}}^{(0)} = \mathbf{0}$ and updates on misclassified points.

Growth of Alignment with Optimal Separator

- For a misclassified point $\overline{\mathbf{x}}_{k+1}, t_{k+1}$:

$$\begin{aligned}\overline{\mathbf{w}}^{(k+1)T}\overline{\mathbf{w}}^* &= \overline{\mathbf{w}}^{(k)T}\overline{\mathbf{w}}^* + \eta\,\overline{\mathbf{x}}_{k+1}^T\overline{\mathbf{w}}^* t_{k+1} \\ &\geq \overline{\mathbf{w}}^{(k)T}\overline{\mathbf{w}}^* + \eta\gamma\end{aligned}$$

- After T updates: $\overline{\mathbf{w}}^{(T)T}\overline{\mathbf{w}}^* \geq T\eta\gamma$.

Controlled Growth of Weight Vector Norm

- Squared norm update:

$$\begin{aligned}\|\overline{\mathbf{w}}^{(k+1)}\|^2 &= \|\overline{\mathbf{w}}^{(k)}\|^2 + 2\eta\,\overline{\mathbf{w}}^{(k)T}\overline{\mathbf{x}}_{k+1}t_{k+1} + \eta^2\|\overline{\mathbf{x}}_{k+1}\|^2 \\ &\leq \|\overline{\mathbf{w}}^{(k)}\|^2 + \eta^2 R^2 \quad (\text{since } \overline{\mathbf{w}}^{(k)T}\overline{\mathbf{x}}_{k+1}t_{k+1} < 0)\end{aligned}$$

- After k updates: $\|\overline{\mathbf{w}}^{(k)}\|^2 \leq k\eta^2 R^2$, so $\|\overline{\mathbf{w}}^{(k)}\| \leq \sqrt{k}\eta R$.

Combining Bounds

- Cauchy-Schwarz: $\overline{\mathbf{w}}^{(k)T}\overline{\mathbf{w}}^* \leq \|\overline{\mathbf{w}}^{(k)}\|\,\|\overline{\mathbf{w}}^*\| = \|\overline{\mathbf{w}}^{(k)}\|$.
- Substituting bounds: $k\eta\gamma \leq \|\overline{\mathbf{w}}^{(k)}\| \leq \sqrt{k}\eta R$.
- Squaring: $k^2\eta^2\gamma^2 \leq k\eta^2 R^2$.
- Dividing by $k\eta^2$ (for $k > 0$):

$$k \leq \left(\frac{R}{\gamma}\right)^2 \tag{1.17}$$

- Number of updates bounded by $(R/\gamma)^2$.
- Bound independent of dimensionality and dataset size.
- Larger margin $\gamma \Rightarrow$ faster convergence.
- If data not linearly separable, algorithm may not converge (oscillates).

Summary

- Perceptron is simple linear classifier with hard threshold.
- Learning by updating weights on misclassified points.
- Guaranteed finite-time convergence for linearly separable data.
- Convergence bound depends only on margin and data radius.
- Limitation: requires linear separability; motivates margin-based methods (SVMs) and nonlinear extensions (neural networks).